Cooperative Search Using Agents for Cardinality Constrained Portfolio Selection Problem

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***Abstract*—This paper presents an agent-based model to select an investment portfolio with a restriction on the number of stocks in it. Daily movements of all the stocks in the market for the past few years are assumed to be available. The scheme deploys a federally structured consortium of agents in the stock market at the start of the historical period. Each agent starts with a pseudorandom portfolio and follows individual investment strategies as it walks through the past data. The agents are designed to emulate some of the characteristics of human investors—adjusting the weights of the stocks based on its own attitude toward risk, occasionally dropping and adding stocks to the portfolio, etc. Periodically, the agents share information about their performances and can switch portfolios. A final cardinality constrained portfolio is constructed by consolidating individual portfolios arrived at by the agents working on the historical data of the stocks. When tested in real markets of the U.K. and Japan, the model suggested portfolios that were quite competitive to, and frequently better than, the portfolios suggested by the mean-variance models.**

***Index Terms*—Agent-based systems, cooperative search, portfolio selection, social learning.**

I. Introduction

MULTIAGENT systems belong to a relatively new genre of systems to solve complex real life problems. A set of autonomous, proactive, intelligent computer programs, which are called agents, share the responsibility of solving the problem. Each agent has its own share of tasks. The agents react to the environment and collaborate with each other to achieve the stated objective. Multiagent systems have been suggested for tackling problems in various disciplines ranging from dynamic job shop scheduling [1] to managing abnormal situations in artificially gas-lifted wells [2] to improving synergetic interactions among a fleet of combat ships [3], to name a few. A number of multiagent systems have also been successfully applied in solving finance-related problems like the stock trading and portfolio management problem (see, e.g., [4] and [5]). Sycara *et al.* [6], [7] proposed design of a portfolio management system that focuses on portfolio monitoring rather than portfolio selection. Portfolio monitoring involves many concurrent goals like monitoring an asset currently held or monitoring buying-and-selling of an asset. Vahidov and Fazlollahi [8] developed a framework for pluralistic multiagent decision support system and tested it in a prototype that would suggest a fixed-size investment portfolio using agents of the "design" and "choice" teams. Parkes and Huberman [9] investigated the potential of using multiagent system for the general portfolio selection. This paper extends Parkes and Huberman's approach to construct a cardinality constrained portfolio and tries it out in real market.

The problem discussed in this paper can be described as follows. An investor wants to invest some amount of money in the stock market. Since there are too many stocks in the market, she has decided to invest only in a fixed number of stocks. She now wants to decide 1) which stocks to invest in, and 2) how to apportion the available money among the stocks chosen. To help the decision-making process, she has access to the daily movements of all the stocks in the market for the past few years. Obviously, her goal is to choose a portfolio that would maximize her expected return—subject to a "reasonable" risk—if she holds on to the portfolio for some period of time in future.

The problem faced by the investor is what is known as the cardinality constrained portfolio selection problem. This is a well-researched problem in the finance arena and can be solved using the widely studied mean-variance (MV) model that, in the instant problem, results in a quadratic mixed integer programming (QMIP) problem and is known to be computationally challenging [10], [11]. Again, this model-based approach assumes a statistical model of stock price dynamics [12] and is as good as the accuracy of the underlying stock market model. These models also presuppose that the investor knows, in quantitative terms, the levels of the expected return, or the risk.

This paper presents an alternate way of addressing the problem using agents. The proposed scheme is a model-free approach that does not make any assumption about the underlying stock price dynamics. Agents of a federally structured consortium emulate some characteristics of human investors like adjusting weights of stocks in the portfolio based on market movements, adding and dropping of stocks from the portfolio, and listening to suggestions from other investors. The agents are assumed to have started operating in the market in the first period of the historical data. Each agent has its own outlook on market volatility and starts with a different pseudorandom portfolio of given cardinality. As stock prices change with the passage of time in the historical period, each agent adjusts the

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weights in its portfolio based on its own perception about the sustainability of the current trend. At fixed intervals, an agent compares the "potential" of the stocks in the market against the "relative goodness" of the stocks in the agent's portfolio and may drop some stocks and include others. The agents use two measures to judge the potential and the relative goodness of the stocks. The proposed measures are intuitive in nature and essentially depend on the trends of the stocks. Every agent periodically broadcasts its performance. A poor performer *may* decide to switch over to the portfolio of the agent performing best at that point in time. At the end of the historical period, the consortium meets and the best performing portfolios are pulled together to suggest one single portfolio of given cardinality to be invested in by the investor.

Each agent in our model is a bounded rational actor as its decision-making capability is seriously compromised because of its bounded cognitive ability, as well as the complexity of the environment in which it operates [13]. All agents receive the same information about the movements of stocks in the market, but each react differently based on its current portfolio and its perception about market volatility. Agents try to overcome individual shortcomings through cooperation among agents having similar goals. They periodically exchange information on portfolio compositions and payoffs, and an agent may decide to imitate the portfolio of another agent performing better at that point in time. Although the two agents will have the same portfolio at the time of switching, each would be following a different replacement strategy and will have a different outlook on future. Thus, the agent-based model that is proposed here is related to the model of *social learning* since the agents base their decisions, at least in part, on the experience of other agents [14]. Considering the massive pool of investors outside the consortium of agents, the stock price dynamics is assumed to be independent of the investment decisions of the agents in the consortium.

The agents of the proposed system pass the test of "agenthood" suggested by Huhns and Singh [15]. Addition of one more agent expands the space for collaboration and may lead to a different portfolio. Once initiated, the agents are autonomous, proactive, and have "intelligence." They react to the changing stock market and, periodically, interact with other agents. The agents in our system work on the past movements of the stock market and do not deal with the current dynamics of it.

The agent-based scheme was tried out on data drawn from FTSE 100 and Nikkei 225 datasets. These are the share indices of the highly capitalized stocks listed on London stock exchange and Japan stock exchange, respectively. The performance of the portfolio suggested by the proposed scheme was found to be quite competitive to, and frequently better than, that of the portfolio suggested by the MV approach when applied on same out-of-sample data.

The rest of this paper is structured as follows. Section II briefly reviews the literature on cardinality constrained portfolio selection problem and introduces the benchmark for comparing with the performance of the proposed system. Section III defines the terminology used in the agent-based model. Section IV details the general scheme of the proposed model. Section V presents the findings of our experiments in real markets. Section VI concludes the paper with directions for future research.

II. Cardinality Constrained Portfolio Selection Problem

The return of an investment portfolio is a random variable which depends upon the choice of the portfolio. Let us assume that the market (the universe) has *n* stocks and let w_i (≥ 0) be the proportion of capital invested in stock *i*. To represent a portfolio, the weights $(w_1, w_2, \ldots, w_n)$ must satisfy a set of constraints that form a feasible set $\mathbf{R}$ of decision vectors. The simplest way to define a feasible set is by the requirement that weights must add up to one. For this basic version of the problem, the set of feasible decision vectors is

$$\mathbf{R} = \left\{ (w_1, \ldots, w_n) : \sum_{i-1}^{n} w_i = 1,\ w_i \geq 0 \quad \forall i = 1...n \right\}.$$

The portfolio selection problem requires models to choose among random variables (portfolios). The purpose of such models is to specify a preference relation among the random variables and to identify random variables that are nondominated with respect to that preference relation.

A very popular decision model for this purpose is the mean-risk model. In a mean-risk model, portfolio decisions are based on two statistics of the portfolio return distribution: the expected return and the value of a risk measure. Markowitz [16] used variance as the risk measure and his work laid the foundation of the present day Modern Portfolio Theory. Although variance as a risk measure has been criticized because of its symmetric nature, and MV model has been criticized for not considering the uncertainty in the estimate of the expected returns [17], "... MV preferences have been extensively used to model the behavior of economic agents choosing among uncertain prospects and have become one of the workhorses of portfolio selection theory" [18] because of its analytical tractability and intuitive interpretation. In this paper, we will be using the MV model as a benchmark to compare the performance of the agent-based model.

The MV formulation with no cardinality constraint results in a quadratic programming (QP) problem and a number of algorithms are available to solve such QP problems [11]. In the MV framework, due to modeling requirements, cardinality restriction on the number of stocks to be held in the portfolio warrants threshold restrictions on the weights of stocks in the portfolio. The resulting problem is a QMIP problem and is difficult to be solved [10], [11].

One formulation of the cardinality constrained MV model is given as follows:

$$\min \sum_{i=1}^{n} \sum_{j=1}^{n} w_i \sigma_{ij} w_j$$

$$\text{subject to}$$

$$\sum_{i=1}^{n} w_i \mu_i \geq d$$

$$\alpha \delta_i \leq w_i \leq \beta \delta_i \forall i = 1, ..., n$$

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$$\sum_{i=1}^{n} \delta_i = k$$

$$\sum_{i=1}^{n} w_i = 1$$

$$w_i \geq 0 \quad \forall i = 1, ..., n$$

$$\delta_i \text{ binary} \quad \forall i = 1, ..., n$$

where σ_{ij} is the covariance between returns of stocks *i* and *j*, μ_i is the expected return of stock *i*, α and β represent the lower and upper thresholds, respectively, on stock weight, *d* is the desired level of expected return, and *k* is the number of stocks to be held in the portfolio. Another formulation of MV model can be maximization of portfolio's expected return with a constraint on portfolio's risk. We intend to compare the proposed model with both the versions of the MV model. Jobst *et al.* [10] use integer restart and reoptimization heuristics for the QMIP when considering MV models with cardinality, threshold, and round lot constraints; they show that the MV efficient frontier, in the presence of these constraints, becomes discontinuous. Bienstock [19] and Lee and Mitchell [20] use QMIP techniques, which can be used to solve the portfolio selection problem with an upper limit on the size of the portfolios. Chang *et al.* [21] use heuristic algorithms—genetic algorithm, tabu search, and simulated annealing—to solve cardinality constrained problems with specified portfolio sizes. This study is motivated by Parkes and Huberman's [9] work on use of multiagent models for general portfolio selection problem and extends the work to solve the cardinality constrained multiperiod portfolio selection problem.

Parkes and Huberman experimented with three types of agents: agents following nonadaptive search, agents following adaptive search, and agents following cooperative search. The agents in nonadaptive search stick to their initial portfolio, buying and selling the stocks as needed to maintain the investment in any stock equal to the initial proportion, throughout the investment period.

Agents of the second type follow adaptive search and use a computationally efficient portfolio updating rule, which is called the χ^2 rule. The χ^2 rule updates the weight of a stock in the agent's portfolio according to the stock's performance in the most recent time period relative to the portfolio's performance and its learning rate η. The learning rates of different agents in the group are different.

The third type of agents follow cooperative search technique. The agents in this group are identical to the agents in the group following adaptive search so far as learning is concerned. In addition, cooperative agents also exchange hints with other agents about its portfolio's recent performance. It can also switch probabilistically to the best performing portfolio among the agents.

The overall portfolio $\boldsymbol{w}_e^t$ of a system of agents in round *t* is given as

$$\boldsymbol{w}_e^t = \sum_{i=1}^{M} \left(\frac{\text{wealth}_i^t}{\sum_{j=1}^{M} \text{wealth}_j^t} \right) \boldsymbol{w}_i^t$$

where wealth_i^t is the wealth of agent *i* at time period *t*, and *M* is the number of agents.

As a prelude to the work reported here, Parkes and Huberman's [9] scheme *per se* was applied on real markets. Parkes and Huberman's model does not constrain the agents to hold a specific number of stocks in their portfolios. An agent's portfolio can have nonzero investments in any number of stocks so long as the investments add up to one. Parkes and Huberman's model, for all the three types of agents, was tried out on the four FTSE 100 and Nikkei 225 datasets detailed in Section V. As suggested by Parkes and Huberman, the initial random portfolio for each agent was obtained from a Dirichlet distribution. The agents followed Parkes and Huberman's approach of portfolio selection, updating, and cooperation. Each of the three types of agents was tried out 100 times on the same data with different initial random portfolios. The final portfolios arrived at by agents following nonadaptive and adaptive search contained *all* the stocks of the respective universe in each of the 100 trials in each dataset. As regards agents following cooperative search, the final suggested portfolio contained 85% or more stocks of the respective universe of stocks in all the 100 trails on two of the datasets. In the remaining two datasets, final portfolio contained more than 85% of the stocks 98 and 85 times, respectively, out of the 100 trials on each dataset. If the asset universe is large, this would imply that an investor, with limited wealth, will have a very small investment across almost all the stocks in the universe. In reality, an investor would prefer to split her investment among a chosen few stocks so that she can track these stocks more closely. Thus, if an investor is interested in limiting her portfolio to say, 10–15% of the stocks in the universe, the Parkes and Huberman model does not serve the purpose of stock selection well.

Here, we propose a cooperative agent-based scheme that can serve as an alternative to solve the cardinality constrained portfolio selection problem. The performance of the proposed model in real markets was found to be quite encouraging.

III. Some Definitions

Let us consider a market with $t = 1, \ldots, T$ discrete investment periods. Let x_i^t denote the price relative of stock *i* in period *t*, the ratio of closing price on day *t* to closing price on day *t* – 1. The vector $\boldsymbol{w} = (w_1, \ldots, w_n)$, where $w_i \geq 0$ and $\sum_{i=1}^{n} w_i = 1$, defines a portfolio; w_i is the fraction of total investment in stock *i*. The return-on-investment in a single period, for an agent with portfolio $\boldsymbol{w}$ and price relatives $\boldsymbol{x} = (x_1, \ldots, x_n)$, is given by the weighted sum over all stocks, i.e., $\boldsymbol{w}.\boldsymbol{x} = \sum_{i=1}^{n} w_i x_i$. The return on investment over *T* periods is given as the product of single period returns $\Re = \prod_{t=1}^{T} \boldsymbol{w}^t.\boldsymbol{x}^t$. The goal of multiperiod portfolio selection is to select, through a sequence of intermediate portfolios, the final portfolio $\{\boldsymbol{w}^T\}$, which would maximize a measure of performance over the final return-on-investment. It may be noted that we are interested only in the end-of-investment period portfolio ignoring the intermediate portfolios. The investment period, in our case, is the historical period that serves the purpose of learning, based on which the investor would take her portfolio decision.

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In the agent-based scheme, the agents should follow a strategy that is strongly competitive with the best offline constant rebalanced portfolio [22]. Parkes and Huberman [9] in their work used the χ^2 rule, which is an approximation to a model-free portfolio selection rule that is strongly competitive with the best offline constant rebalanced portfolio. Hence, we adopt the same rule in our work. In the following, we revisit the definitions for χ^2 *rule* and performance measure, as proposed in [9]. We also define two other measures used by the agents in our scheme.

Definition 1 (The χ^2 portfolio updating rule): If $\boldsymbol{w}^t$ represents the portfolio in investment period *t* and $\boldsymbol{x}^t$ gives the price relatives observed in that investment period, the portfolio in period $t+1$, represented as $\boldsymbol{w}^{t+1}$, is given as

$$w_i^{t+1} = w_i^t \left(\eta \left(\frac{x_i^t}{\boldsymbol{w}^t . \boldsymbol{x}^t} - 1 \right) + 1 \right)$$

where $x_i{}^t = P_i{}^t / P_i{}^{t-1}$ is the price relative of stock *i* at time period *t* with respect to the previous time period, $P_i{}^t$ being the price of stock *i* at time *t*; and $\eta > 0$ is the learning rate.

This rule increases the fraction of investment in stocks that perform better than the portfolio, while decreasing the fraction of wealth invested in the underperforming stocks. Learning parameter η in the χ^2 rule differs across agents and determines an agent's responsiveness to the changes in stock prices, as conveyed by the price relatives. It adjusts weight of stocks moving in a direction that will give a better performance if the price relatives in the current period characterize future periods.

Definition 2 (Perf, Performance Measure): The performance, *Perf,* of a multiperiod portfolio $\{\boldsymbol{w}^T\}$for an agent with a logarithmic utility for return on investment is

$$Perf = E_{\{x^T\}} \left[\log \left(\prod_{t=1}^{T} \boldsymbol{w}^t . \boldsymbol{x}^t \right) \right]$$

where the expectation is taken over sequence of price relatives, i.e., $\{x^T\}$, distributed according to market price dynamics, and *T* is the number of investment periods. Parkes and Huberman provide detailed justification for using the logarithmic utility and concludes that "a logarithmic utility function represents the preferences of a risk-averse investor and is useful because it allows tractable analysis" [9].

Definition 3 (Gmeasure$_i$, Global Measure of Stock): The measure of stock *i*'s overall performance, *Gmeasure$_i$*, is defined as

$$Gmeasure_i = \frac{P_i^t}{P_i^0} \times \frac{c_i}{t}$$

where P_i^t is the price of stock *i* in period *t*, P_i^0 is the price of stock *i* in period 0, c_i is the count of number of times stock *i's* price has risen relative to last day (or, equivalently, price relative is greater than 1), and *t* is the current time period.

Gmeasure$_i$ measures the goodness of stock *i*. The ratio $\frac{P_i^t}{P_i^0}$ of the price of the *i*th stock on day *t* to that on day 0 is an indication of the return that would have been realized if one unit of the stock was purchased at the beginning and held for *t* investment periods. $\frac{c_i}{t}$ is an indication of the pattern the stock price has followed in reaching the price of day *t*. It is the fraction of the time the stock has shown rising tendency. $\frac{P_i^t}{P_i^0}$ shows the gain of the stock, while $\frac{c_i}{t}$ reflects the consistency of the stock. The product is expected to give an indication of the stock's potential.

Definition 4 ($Rmeasure_i^m$, Portfolio Measure of Stock): The measure of stock *i*'s performance in the portfolio of agent *m,* relative to other stocks in the portfolio, $Rmeasure_i^m$ is defined as

$$Rmeasure_i^m = w_i^t \times \frac{P_i^t}{P_i^0} \times \frac{c_i}{t} = w_i^t \times Gmeasure_i$$

where $\boldsymbol{w}^t = (w_1^t, \ldots, w_n^t)$ is the portfolio held by the agent *m* at time *t*.

$Rmeasure_i^m$ tries to capture the relative importance of a stock in agent *m*'s portfolio together with the potential of the stock.

IV. Agent-Based Search Scheme

The agents of an agent-based system must have autonomy, should be capable to reason and act in their environment, and should be able to interact with other agents [23]. Aguilar *et al.* [2] also identify autonomy, mobility, sociability, reactivity, proactivity, and intelligence to be the characteristics of agents in a multiagent system. In our scheme of affairs, each agent emulates a human investor who has his own attitude toward market volatility, characterized by the learning rate of that agent. Each agent has an objective to form an independent cardinality constrained portfolio based on movements of stock prices in the past. Agents have autonomy to decide the composition of the portfolio, or the weights of the stocks in it. Agents react to the dynamics of the stock market and can proactively decide to drop and add stocks to its portfolio. The agents try to improve their performance by exchanging information with other agents. In our model, the agents belong to a consortium of agents, and the portfolios formed by the group of agents are "pulled together" at the end to form the final cardinality constrained portfolio for future investment.

Let *k* be the cardinality of the portfolio to be constructed. At the first investment period of the historical time under consideration, each agent is handed over a pseudorandom portfolio with nonzero investments in *k* stocks, each having same weight in the portfolio. Agents do not have any idea about how the stocks would perform in subsequent time periods. As each agent steps into the next time period, it observes the price movements and adjusts the weights of the stocks. The agent can also periodically replace selected poor-performing stocks with "better" stocks. From time to time, the agents share information among themselves. Based on the information received, an agent has the option to reinitialize its portfolio with the portfolio of some other agent. Thus, the proposed model for portfolio construction is a set of five activities: 1) each agent is assigned an initial pseudorandom portfolio; 2) at every subsequent time period, each agent, based on its own risk perception, updates the weights of the stocks in the portfolio in response to changes in stock prices; 3) each agent periodically replaces a few poor-performing stocks in its portfolio with better performing stocks; 4) at intervals, agents exchange hints among themselves, and a poor-performing agent

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may decide to follow the best performer's strategy; (e) finally, the consortium takes into consideration the performance of all the agents and constructs the final cardinality constrained portfolio. This is the recommended portfolio that the investor can invest in. We are, in essence, talking of two types of agents: coordinator agent and task agents [24]. The coordinator agent initiates the pool of task agents (activity 1 above) and deploys them in the stock market at the beginning of the historical period. It then waits till the task agents complete their job. At the end, it is the coordinator agent's job to construct the final portfolio (activity 5). Each task agent is independently responsible for activities 2, 3, and 4 above. We now discuss each of these activities in detail.

A. Initialization

On the first period of the historical data, every agent is assigned a share of initial wealth $= V_0/M$, where V_0 is the initial total wealth and M is the total number of agents in the group. M initial pseudorandom portfolios are selected and assigned to the agents in the group. An initial portfolio consists of k of stocks having nonzero weights, and $n - k$ stocks with zero weights. These k stocks in an agent's portfolio are assigned equal weight of $1/k$, i.e., each stock in each agent's portfolio has equal proportion of wealth to begin with. For each agent, the k stocks to be assigned nonzero weights are selected in such a way that each stock is present in approximately equal number of agents' portfolio across the group.

A global list of all the stocks is maintained and the first agent is free to pick up k number of stocks randomly from this global list. These are the only stocks that will have nonzero weights in the portfolio of the first agent. The stocks that are already selected are removed from the global list. The second agent then randomly picks up k number of stocks from the remaining stocks in the global list. These stocks are also removed from the list reducing the choices available to the next agent. The process is continued till there are less than or equal to k number of stocks left in the global list of stocks. The stocks left in the global list are then included in the next agent's set of stocks. Once the global list is fully exhausted, the list is fully replenished with all the stocks, and any shortfall in the last assigned agent's portfolio is made up randomly from the fully replenished global list (and the selected stocks get omitted from the global list for next agent). The process is repeated till all the agents receive their initial pseudorandom portfolios.

The aforementioned scheme for deciding the initial portfolios deserves some discussion. As known from other models for portfolio selection in finance like the MV model, the risk and return of the portfolio is not just a function of individual stock's performance but also depend upon the way stocks behave together. For example, the covariance of stocks in the portfolio is also a contributing factor to the risk in the MV model. By assigning the stocks in the way discussed previously, we try to capture this effect. While a stock is assigned to approximately the same number of agents across the group, the other stocks it coexists in different agents' portfolios are different. We, hence, get an indication of the way a portfolio performs with different combination of stocks and try to make the set of choices more balanced and heterogeneous.

During initialization, each agent is also assigned a learning rate η uniformly distributed between lower bound η_i and upper bound η_u. The learning rate characterizes the risk appetite of an agent as explained in the following section.

B. Updating

As agents step into the next time period t, they come to know the new prices of the stocks in their respective portfolios. Each agent calculates the recent price relatives $x_i{}^t$ $(= P_i{}^t / P_i{}^{t-1})$ of the stocks in its portfolio. The agent will now use the χ^2 *portfolio updating rule* to adjust the weights of the stocks in its portfolio based on discovered price relatives and the agent's learning rate η that acts as a surrogate of the agent's perception about market volatility. The updating rule ensures that the weights in the updated portfolio add up to one. A stock with a price relative greater than 1 will definitely see an increase in its weight in the portfolio, but the amount of increase will depend on the price relatives of other stocks in the portfolio and the agent's η. The agents have heterogeneous learning rates η. Some agents will have more confidence on the sustainability of the current price relative in future and will be more aggressive in adjusting the weights of the stocks. Such agents are characterized by high learning rates. On the other hand, an agent with a relatively low learning rate is conservative and would opt for small adjustments in the relative weights of stocks in the portfolio. This step ensures continuous adjustment of agent's portfolio normally desired for good performance.

C. Replacement

The updating process, as described previously, adjusts relative weights of the k stocks in an agent's portfolio. However, some stocks in the portfolio may turn out to be consistent poor performers relative to others, and hence, its weight will slide at a rate that would depend on the learning rate of the agent. The replacement activity allows an agent to periodically get rid of poor-performing stocks, while ensuring that, at any point in time, the portfolio has exactly k stocks with nonzero investment.

An agent keeps a count of the time periods for which investment in the same set of stocks was being maintained (time since last change in portfolio's composition). After a prespecified *replacement window* of size λ time periods, the agent identifies relatively poor performers and replaces them. The *Portfolio measure of stock*, $Rmeasure_i^m$, gives an intuitive measure of stock i*'s* goodness vis-à-vis other stocks in the portfolio of agent m. After every replacement window size, y stocks in the agent's current portfolio with lowest value of the $Rmeasure_i^m$ get replaced with other better performing stocks from the asset universe.

The stock that would replace an outgoing stock of the portfolio is selected as follows. A global list of all the stocks, sorted in descending order of *Global measure of stock, Gmeasure*$_i$, is maintained in each time period. For any outgoing stock from an agent's portfolio, all the stocks not already present in that agent's portfolio and having *Gmeasure*$_i$ values greater than that

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of the outgoing stock form a set of eligible incoming stocks. The replacing stock is then picked up randomly from the set of these eligible stocks. If the set of eligible stocks is empty, the current outgoing stock is not replaced. The replacing stock enters the current portfolio with the same weight as the outgoing stock.

At the end of every λ periods, every agent first carries out replacement of stocks in its portfolio and, then, follows the χ^2 portfolio *updating rule* to update the weights of stock in the *new* portfolio.

D. Cooperation

In the previous steps, agents have been taking the portfolio decisions in isolation. However, agents can possibly improve their performances by sharing information with other agents. In the proposed model, the agents exchange hints among themselves to indicate the performance of their portfolios. An agent performing poorly may switch to the portfolio of the agent performing best at that time. Both agents have the same portfolio immediately after switching. However, they follow separate strategies as represented by their different learning rates and their subsequent stock replacements, which are likely to be different. Additional parameters, i.e., the switching probability p and exchange window size τ, decide the extent of cooperation among the agents.

Every agent maintains a count of the periods for which it is holding the current portfolio since last portfolio switching. When this count is greater than or equal to τ, the agent broadcasts its average performance and its current portfolio. The average performance is computed as the average per-period return over the finite number of recent periods τ. An agent performing poorly can switch, with probability p, to the best performing portfolio received over broadcast. If the switch takes place, the period count for which the current portfolio is held is reset to zero.

If the time period of switch and replacement coincides, an agent first carries out portfolio switching probabilistically before carrying out replacement of stocks and updating of the portfolio.

A point to note is that replacement and cooperation activities try to achieve periodic changes in the composition of the portfolio. Changing the composition on a daily basis would possibly not be advisable since a stock's potential can best be judged based on its performance over a stretch of time periods.

E. Portfolio Construction

Each agent carries on the activities of updating, replacement, and cooperation till it reaches the end of the historical periods under consideration and reports its own cardinality constrained portfolio to the consortium. The last step in our model is the construction of the final portfolio with nonzero investment in only k of stocks.

The portfolios reported by the agents are arranged in descending order of their end-of-investment period wealth. The portfolios of the top q agents are taken into consideration for the purpose of final portfolio formation. Each agent, out of these top q agents, recommends top v stocks from its portfolio. The weights in stock i are added across the recommended stocks to get w_i which, in a sense, represents the importance given to stock i by the best performing agents. The portfolio so formed will, in general, have more than k number of stocks, and the sum of final weights across all stocks in this portfolio will be more than one. The final cardinality constrained portfolio is formed by picking up k number of stocks with highest weights from this candidate portfolio and normalizing them so that the weights add up to one.

In our experiments, the final portfolio is formed by averaging across many trials. Across trials, the averages of the weights of stocks in top q agents in every trial are considered, and then, the aforementioned procedure is applied.

TABLE I
DATASET CHARACTERISTICS

Dataset name	Dataset Id	In-sample period	No of periods	No of stocks	Out-of-Sample periods		
					OS1	OS2	OS3
FTSE 100	F1	Jan 01 - Dec 04	1012	82	Jan 05	Jan-Feb 05	Jan-Mar 05
FTSE 100	F2	Jan 04 - Dec 07	1011	89	Jan 08	Jan-Feb 08	Jan-Mar 08
Nikkei 225	N1	Jan 01 - Dec 04	983	192	Jan 05	Jan-Feb 05	Jan-Mar 05
Nikkei 225	N2	Jan 04 - Dec 07	984	207	Jan 08	Jan-Feb 08	Jan-Mar 08

V. COMPUTATIONAL EXPERIENCE

In this section, we investigate the performance of the proposed model in real markets. For this, we chose one European and one Asian market, and for each, we considered one bullish and one bearish period. For all the four cases, the historical period was taken to be four years. We check the out-of-sample performance of the portfolios constructed using the agent-based approach and compare these with the performance of the portfolios formed using the traditional MV approach.

A. Dataset Description

Four datasets, two from FTSE 100 and two from Nikkei 225 over the same periods, were used for this investigation. The characteristics of the datasets are given in Table I. As evident from Table I, for the first dataset, daily price relatives of 82 stocks that belong to the FTSE100 index throughout the period Jan 2001–December 2004 were considered. For each of the remaining 18 stocks' data, there is at least one missing data item in the specified period. For the out-of-sample analysis, we considered three out-of-sample periods of different time spans: Jan 2005, Jan-Feb 2005, and Jan-March 2005 spanning one, two, and three months, respectively. The other three in-sample periods, and corresponding out-of-sample periods, were decided in a similar manner.

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TABLE II
PARAMETERS' VALUES FOR DIFFERENT DATASETS

Dataset	Trials	M	k	λ	τ	η_l	η_u	p	y	q	v
F1	100	300	15	100	120	0.1	0.2	0.01	5	30	5
F2	100	300	15	75	120	0.1	0.2	0.01	5	30	10
N1	100	240	25	75	100	0.1	0.2	0.01	8	30	10
N2	100	240	25	22	100	0.8	0.9	0.01	5	20	5

TABLE III
PERFORMANCE ON DATASETS

Dataset	Period	Statistics	Multi-agent	MV-1	MV-2	FTSE100/ Nikkei225
F1	In-sample	Exp Ret	0.0011	0.0011	0.0012	-0.0003
		Std Dev	0.0102	0.0087	0.0102	0.0130
	OS1	Avg Ret	0.0013	0.0007	0.0012	0.0004
	OS2	Avg Ret	0.0019	0.0015	0.0017	0.0008
	OS3	Avg Ret	0.0006	0.0004	0.0003	0.0003
F2	In-sample	Exp Ret	0.0014	0.0014	0.0016	0.0004
		Std Dev	0.0128	0.0104	0.0128	0.0080
	OS1	Avg Ret	-0.0039	-0.0039	-0.0039	-0.0043
	OS2	Avg Ret	-0.0012	-0.0029	-0.0030	-0.0022
	OS3	Avg Ret	-0.0016	-0.0024	-0.0027	-0.0020
N1	In-sample	Exp Ret	0.0010	0.0010	0.0014	-0.0002
		Std Dev	0.0174	0.0129	0.0174	0.0154
	OS1	Avg Ret	0.0027	0.0013	0.0018	-0.0005
	OS2	Avg Ret	0.0027	0.0015	0.0020	0.0006
	OS3	Avg Ret	0.0017	0.0013	0.0017	0.0003
N2	In-sample	Exp Ret	0.0019	0.0019	0.0022	0.0004
		Std Dev	0.0191	0.0153	0.0191	0.0111
	OS1	Avg Ret	-0.0059	-0.0053	-0.0055	-0.0063
	OS2	Avg Ret	-0.0018	-0.0019	-0.0013	-0.0030
	OS3	Avg Ret	-0.0026	-0.0021	-0.0015	-0.0034

B. Parameter Values

The scheme requires preselection of quite a few parameters: the replacement window λ, the exchange window τ, the switching probability p, the number q of best performing agents, the number y of stocks to be replaced, the number v of stocks to be recommended by the best performers, and the range of learning rates (η_l, η_u) to choose from. How do we choose optimal values of these parameters so that we get a portfolio that would perform "well" in future? A formal method for deciding the parameter values entails detailed investigation and analysis into the properties of these parameters. However, we use an empirical approach to select parameter values. Each of our datasets has 48 months of data. We divide this dataset into two parts: Part A, consisting of first 47 months of data, and Part B, containing only the last (most recent) month's data. We now treat the Part A data as if it is the only in-sample data available to us and consider Part B to be our out-of-sample period. We then run our agent-based model on Part A and check its performance on Part B. We keep on tuning the parameters till we get the set of parameter values that gives the "best" average return on Part B. The portfolio, thus, arrived at using Part A data is the portfolio that is tested on actual out-of-sample dataset. The values of the parameters, thus, obtained for the four datasets are given in Table II.

There were 100 trials over each in-sample period, the outcomes of which were used to arrive at the final portfolio as discussed earlier. The cardinality k of the chosen portfolio was considered 15 and 25, respectively, for FTSE and Nikkei datasets. The number of agents used was 300 for the FTSE datasets and 240 for the Nikkei datasets. The number of agents had to be reduced for the Nikkei datasets, since the model became very sluggish due to substantially more number of stocks in the universe.

C. Portfolio Performance

The performance of the portfolios constructed by the proposed model vis-à-vis the corresponding index, and the MV models over the chosen periods are summarized in Table III. The suggested portfolios have higher in-sample expected return and comparable risk when compared with the corresponding index (FTSE 100, Nikkei 225) during the same period. When tested on out-of-sample data, in all the cases across different datasets, the agent-based portfolios achieve higher average returns than the corresponding index.

We were more interested in comparing the agent-based portfolios' performance with the performance of the portfolios constructed using traditional cardinality constrained MV models (with threshold restrictions set at 0.00001 and 1). We considered both the possible versions of the MV models.

1) MV1: The portfolio is obtained by minimizing portfolio's variance with the portfolio's expected return constrained to be greater than or equal to agent-based portfolio's in-sample expected return.
2) MV2: The portfolio is obtained by maximizing portfolio's expected return with the portfolio's variance constrained to be less than or equal to agent-based portfolio's in-sample variance.

It is interesting to note that, going by in-sample results, the MV models provide better promising portfolios. In all the datasets, the in-sample portfolios of MV1 and MV2 outperform the portfolio of agent-based model in risk and return, respectively. It is worth pointing out that the portfolio of either of the MV models is formed at the end of the historical period with *hindsight*, i.e., with all the information about the stock price changes during the entire historical period. The portfolio, thus, suggested would have been the best portfolio to be held during in-sample period. In contrast, the agent-based model builds

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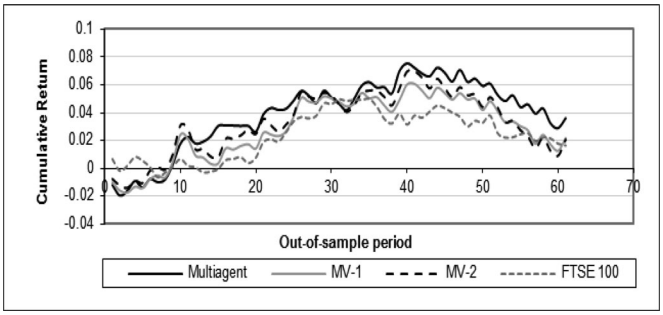

Fig. 1. Out-of-sample performance on F1.

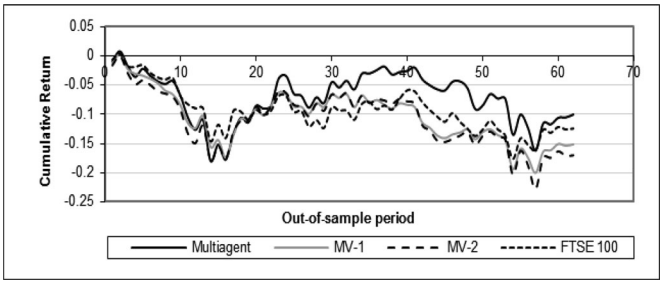

Fig. 2. Out-of-sample performance on F2.

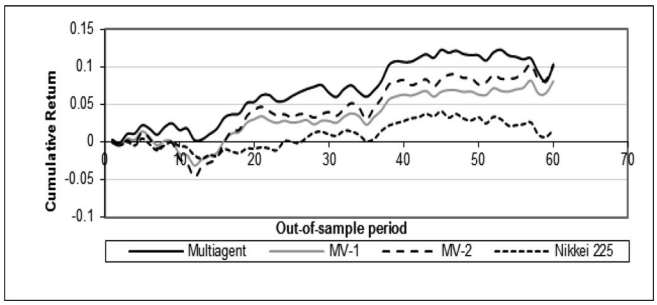

Fig. 3. Out-of-sample performance on N1.

the final portfolio in an iterative fashion, adjusting yesterday's portfolio based on price movements of today. However, when it comes to out-of-sample performance on holding periods of one, two, or three months, the portfolio formed by the agent-based model has always ensured better average return than MV1 except two cases in the last dataset. Compared with MV2, the agent-based portfolio again provides better average return in three out of the four datasets. The daily out-of-sample cumulative returns that are depicted in Figs. 1–4 also highlight better performance of the agent-based portfolios vis-à-vis MV portfolios in three datasets.

VI. Conclusion

It is interesting to conjecture why the agent-based model does better than the MV models in most of the out-of-sample cases. One possible reason could be the agent-based model's ability to capture the stock price trend better, as reflected in the model's portfolio updating and stock replacement activities. The portfolio updating activity ensures that a stock doing relatively better than the other stocks in the portfolio is progressively

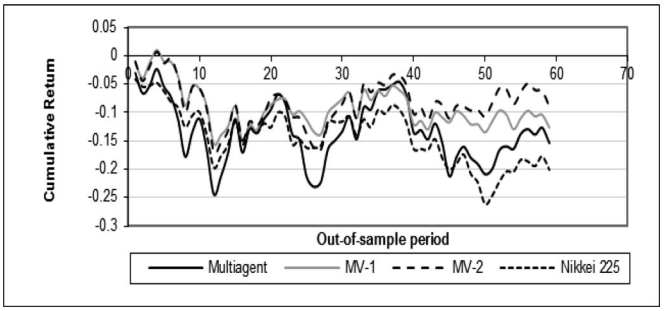

Fig. 4. Out-of-sample performance on N2.

allocated a larger proportion of money while doing the opposite for the bad stock. The stock replacement activity makes use of *Rmeasure* and *Gmeasure* that act as surrogates of the trends of the stocks. The replacement step tries to replace a bad stock in the portfolio, with a stock having price gains more number of times, as well as having potential for higher return over the holding period.

The results indicate that the agent-based approach has potential. However, a lot of investigation needs to be done before we arrive at a robust scheme. What would be a generic prescription for choosing the parameter values? How sensitive is the performance of the portfolio to the choice of parameter values? The current scheme of cooperation is a broadcast mechanism. Would it be better to devise a peer-to-peer dialogue and exchange? We leave these issues as agenda for future research.

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